
HPI & Life Expectancy Project — Data Cleaning Script

Datasets:

1. HPI 3 — hpi_3_complete_file.xlsx (8,057 CA census tracts)
2. USALEEP — CDC life expectancy at birth by tract (to be joined)

Goal: Clean and merge data, prep for LASSO / Random Forest / XGBoost models

```
set.seed(42)

library(tidyverse)
library(readxl)
library(rsample)
library(janitor)    # clean_names(), tabyl()
library(skimr)      # skim() for quick EDA summaries
```

1. LOAD DATA

```
path<- "~/Documents/ChicoState/Spring26/M485/project/hpi_final/"
hpi_raw    <- read_excel("hpi_3_complete_file.xlsx", sheet = "HPI 3")
usaleep_raw <- read_excel("CA_A.CSV.xlsx")
```

2. STANDARDIZE COLUMN NAMES

```
hpi <- hpi_raw %>%
  clean_names()           # snake_case, removes spaces/special chars

usaleep <- usaleep_raw %>%
  clean_names()           # "Tract ID" -> tract_id, "e(0)" -> e_0_, etc.
```

3. INSPECT & DIAGNOSE

```
glimpse(hpi)
```

```
## Rows: 8,057
## Columns: 84
## $ geo_id      <dbl> 6001400100, 6001400200, 6001400300, 6001400400~
## $ name        <chr> "Census Tract 4001, Alameda County, California~
## $ county      <chr> "Alameda", "Alameda", "Alameda", "Alameda", "A~
## $ pop         <dbl> 3120, 2007, 5051, 4007, 4124, 1745, 5128, 4069~
## $ pctgqtract  <dbl> 0.000000000, 0.046337818, 0.007919224, 0.01247~
## $ urban_type  <chr> "urban_area", "urban_area", "urban_area", "urb~
## $ leb         <dbl> 87.2000, 81.5000, 87.2000, 82.9000, 78.8000, 7~
## $ leb_pctl    <dbl> 0.98254172, 0.64017972, 0.98254172, 0.78883184~
## $ hpi         <dbl> 1.366160697, 1.432718291, 1.178741654, 0.96872~
## $ hpi_pctl    <dbl> 0.99666239, 0.99833119, 0.98793325, 0.96405648~
## $ hpi_quartile <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 3, 2, 2, 2~
## $ hpi_least_healthy_25pct <chr> "No", "No", "No", "No", "No", "No", "No", "No"~
## $ economic    <dbl> 1.770345592, 1.754657559, 1.654709538, 1.22758~
## $ economic_pctl <dbl> 0.98703466, 0.98600770, 0.98241335, 0.93774069~
## $ education   <dbl> 1.3639882, 1.4044192, 1.0491379, 0.6495117, 0.~
## $ education_pctl <dbl> 0.9901155, 0.9933248, 0.9323492, 0.8141207, 0.~
## $ insurance   <dbl> 1.22521065, 1.05534707, 1.33355398, 0.99561639~
## $ insurance_pctl <dbl> 0.9695764, 0.9038511, 0.9946085, 0.8743261, 0.~
## $ clean_enviro <dbl> 1.13263582, 0.68130947, 0.74168714, 0.89097128~
## $ clean_enviro_pctl <dbl> 0.9861361, 0.8421053, 0.8659820, 0.9241335, 0.~
## $ housing     <dbl> 1.04208175, 0.95586848, 0.60915313, 0.32742686~
## $ housing_pctl <dbl> 0.98164313, 0.96854942, 0.84698331, 0.67137356~
## $ neighborhood <dbl> 1.217721452, 0.685531985, -0.342743732, 0.2230~
## $ neighborhood_pctl <dbl> 0.9775353, 0.9342747, 0.2137356, 0.7608472, 0.~
## $ social      <dbl> 1.02386907, 1.21286892, 0.90789689, 0.99361831~
## $ social_pctl  <dbl> 0.88241335, 0.93491656, 0.84069320, 0.87175866~
## $ transportation <dbl> 0.978526728, 1.773992572, 1.305814427, 1.26397~
## $ transportation_pctl <dbl> 0.9750963, 0.9992298, 0.9911425, 0.9890886, 0.~
## $ abovepoverty <dbl> 0.8957728, 0.8936810, 0.8970265, 0.7891838, 0.~
## $ abovepoverty_pctl <dbl> 0.88973042, 0.88575096, 0.89204108, 0.63581515~
## $ automobile   <dbl> 0.975, 0.954, 0.848, 0.906, 0.884, 0.932, 0.85~
## $ automobile_pctl <dbl> 0.722464698, 0.509948652, 0.104364570, 0.23985~
## $ bachelors    <dbl> 0.8668792, 0.8651757, 0.7202366, 0.7591995, 0.~
## $ bachelors_pctl <dbl> 0.9975610, 0.9971759, 0.9469833, 0.9641849, 0.~
## $ censusresponse <dbl> 0.8171195, 0.8592758, 0.7910000, 0.8060257, 0.~
```

```

## $ censusresponse_pctile <dbl> 0.8534018, 0.9431322, 0.7731707, 0.8204108, 0.~
## $ commute <dbl> 0.2526395, 0.4695238, 0.5137552, 0.4197258, 0.~
## $ commute_pctile <dbl> 0.9252888, 0.9756098, 0.9820282, 0.9677792, 0.~
## $ dieselpm <dbl> 0.09288581, 0.59102663, 0.49717906, 0.32702737~
## $ dieselpm_pctile <dbl> 0.657766367, 0.071245186, 0.100641849, 0.20706~
## $ employed <dbl> 0.779, 0.869, 0.882, 0.864, 0.837, 0.863, 0.81~
## $ employed_pctile <dbl> 0.74255456, 0.98626444, 0.99281130, 0.98299101~
## $ h2ocontam <dbl> 110.4124, 110.4124, 110.4124, 110.4124, 110.41~
## $ h2ocontam_pctile <dbl> 0.9754814, 0.9754814, 0.9754814, 0.9754814, 0.~
## $ homeownership <dbl> 0.88940449, 0.61983471, 0.41540352, 0.40954495~
## $ homeownership_pctile <dbl> 0.95109114, 0.56007702, 0.30205392, 0.29614891~
## $ houserepair <dbl> 1.0000000, 1.0000000, 1.0000000, 1.0000000, 0.~
## $ houserepair_pctile <dbl> 0.809435173, 0.809435173, 0.809435173, 0.80943~
## $ inhighschool <dbl> 1.0000000, 1.0000000, 1.0000000, 1.0000000, 1.~
## $ inhighschool_pctile <dbl> 0.63016688, 0.63016688, 0.63016688, 0.63016688~
## $ inpreschool <dbl> 0.8571429, 0.8928571, 0.7876712, 0.4057971, 0.~
## $ inpreschool_pctile <dbl> 0.8576380, 0.8777920, 0.8017972, 0.3514121, 0.~
## $ insured <dbl> 0.9885910, 0.9752560, 0.9970964, 0.9705669, 0.~
## $ insured_pctile <dbl> 0.9695764, 0.9038511, 0.9946085, 0.8743261, 0.~
## $ ownsevere <dbl> 0.12669683, 0.03838384, 0.02790698, 0.12213741~
## $ ownsevere_pctile <dbl> 0.38838254, 0.93851091, 0.96200257, 0.41418485~
## $ ozone <dbl> 0.02937227, 0.02937227, 0.02937227, 0.02937227~
## $ ozone_pctile <dbl> 0.9831836, 0.9831836, 0.9831836, 0.9831836, 0.~
## $ parkaccess <dbl> 1.0000000, 0.9179331, 0.3325797, 0.8422900, 1.~
## $ parkaccess_pctile <dbl> 0.8005135, 0.4711168, 0.1397946, 0.3956354, 0.~
## $ percapitaincome <dbl> 116300, 90742, 80071, 69737, 52438, 58299, 431~
## $ percapitaincome_pctile <dbl> 0.9892169, 0.9672657, 0.9462131, 0.9133504, 0.~
## $ pm25 <dbl> 8.953297, 9.298860, 9.445812, 9.369731, 9.3697~
## $ pm25_pctile <dbl> 0.6397946, 0.5827985, 0.5630295, 0.5741977, 0.~
## $ rentsevere <dbl> 0.00000000, 0.04179104, 0.13718412, 0.16371681~
## $ rentsevere_pctile <dbl> 0.9915918, 0.9716303, 0.8518614, 0.7902439, 0.~
## $ retail <dbl> 0.5838408, 12.7276154, 20.3711296, 8.2620627, ~
## $ retail_pctile <dbl> 0.1812580, 0.8862644, 0.9417202, 0.8012837, 0.~
## $ treecanopy <dbl> 0.36433798, 0.19708371, 0.05838530, 0.11378516~
## $ treecanopy_pctile <dbl> 0.97483954, 0.92528883, 0.57432606, 0.83119384~
## $ uncrowded <dbl> 1.0000000, 0.9929162, 0.9786973, 0.9866815, 0.~
## $ uncrowded_pctile <dbl> 0.9743902, 0.9084724, 0.7758665, 0.8492940, 0.~
## $ voting <dbl> 0.8764185, 0.8772267, 0.8761395, 0.8800317, 0.~
## $ voting_pctile <dbl> 0.85237484, 0.85532734, 0.85096277, 0.86790757~
## $ latino_pct <dbl> 0.038, 0.087, 0.069, 0.121, 0.095, 0.075, 0.19~
## $ white_pct <dbl> 0.743, 0.735, 0.680, 0.637, 0.454, 0.493, 0.38~
## $ black_pct <dbl> 0.034, 0.026, 0.091, 0.066, 0.214, 0.205, 0.28~
## $ asian_pct <dbl> 0.125, 0.085, 0.117, 0.105, 0.108, 0.097, 0.06~
## $ multiple_pct <dbl> 0.060, 0.065, 0.019, 0.062, 0.122, 0.101, 0.07~
## $ native_am_pct <dbl> 0.000, 0.002, 0.000, 0.009, 0.000, 0.002, 0.00~
## $ pacific_isl_pct <dbl> 0.000, 0.000, 0.005, 0.000, 0.005, 0.009, 0.00~
## $ other_pct <dbl> 0.000, 0.000, 0.019, 0.000, 0.001, 0.019, 0.00~
## $ version <dtm> 2022-02-10, 2022-02-10, 2022-02-10, 2022-02-1~
## $ notes <chr> "(c) Copyright 2022 Public Health Alliance of ~

```

```
skim(hpi)
```

Table 1: Data summary

Name	hpi
Number of rows	8057
Number of columns	84
Column type frequency:	
character	5
numeric	78
POSIXct	1
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
name	0	1.00	39	55	0	8057	0
county	0	1.00	4	15	0	58	0
urban_type	1	1.00	5	13	0	4	0
hpi_least_healthy_25pct	267	0.97	2	3	0	2	0
notes	8056	0.00	1446	1446	0	1	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
geo_id	0	1.00	6.054897e+00	3.148362	0.0000e+00	0.37264302	0.05905250	5.073016e+00	6.095041e+09	
pop	0	1.00	4.875700e+04	2280.50	0.0000e+00	0.159.00	4608.00	5.929000e+03	3.875400e+04	
pctgqtract	45	0.99	2.000000e-02	0.09	0.0000e+00	0.00	0.00	1.000000e-01	1.000000e+00	
leb	267	0.97	8.026000e+01	3.39	6.4400e+01	78.00	80.40	8.250000e+01	9.310000e+01	
leb_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
hpi	267	0.97	0.000000e+00	0.58	-	-0.43	0.02	4.300000e-01	1.700000e+00	
hpi_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
hpi_quartile	267	0.97	2.500000e+00	0.12	1.0000e+00	2.00	2.50	3.750000e+00	4.000000e+00	
economic	267	0.97	0.000000e+00	0.84	-	-0.61	0.03	6.000000e-01	2.810000e+00	
economic_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
education	267	0.97	0.000000e+00	0.67	-	-0.45	-0.04	4.800000e-01	1.620000e+00	
education_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
insurance	267	0.97	0.000000e+00	0.00	-	-0.52	0.24	7.500000e-01	1.370000e+00	
insurance_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
clean_enviro	267	0.97	0.000000e+00	0.66	-	-0.54	0.09	5.000000e-01	1.740000e+00	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
clean_enviro_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
housing	267	0.97	0.000000e+00	0.64	-	-0.36	0.09	4.400000e-01	1.360000e+00	
housing_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
neighborhood	267	0.97	0.000000e+00	0.57	-	-0.25	0.03	2.100000e-01	1.207000e+01	
neighborhood_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
social	267	0.97	0.000000e+00	0.89	-	-0.62	0.11	6.900000e-01	1.960000e+00	
social_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
transportation	267	0.97	0.000000e+00	0.39	-	-0.18	0.01	1.500000e-01	1.960000e+00	
transportation_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
abovepoverty	267	0.97	6.900000e-01	0.18	8.0000e-02	0.56	0.72	8.400000e-01	9.900000e-01	
abovepoverty_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
automobile	267	0.97	9.300000e-01	0.08	1.2000e-01	0.91	0.95	9.800000e-01	1.000000e+00	
automobile_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	9.800000e-01	
bachelorsd	267	0.97	3.300000e-01	0.21	0.0000e+00	0.15	0.29	4.900000e-01	9.200000e-01	
bachelorsd_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
censusresponse	267	0.97	7.000000e-01	0.11	1.1000e-01	0.64	0.72	7.800000e-01	9.400000e-01	
censusresponse_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
commute	267	0.97	9.000000e-02	0.12	0.0000e+00	0.02	0.05	1.100000e-01	9.100000e-01	
commute_pctile	267	0.97	5.000000e-01	0.29	1.0000e-02	0.25	0.50	7.500000e-01	1.000000e+00	
dieselpm	267	0.97	2.200000e-01	0.30	0.0000e+00	0.07	0.15	2.900000e-01	1.461000e+01	
dieselpm_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
employed	267	0.97	7.300000e-01	0.08	1.4000e-01	0.68	0.74	7.800000e-01	9.500000e-01	
employed_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
h20contam	267	0.97	4.773600e-02	0.19	3.2570e+03	303.71	430.43	6.783700e+02	1.257750e+03	
h20contam_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
homeownership	267	0.97	5.500000e-01	0.24	0.0000e+00	0.37	0.58	7.500000e-01	9.900000e-01	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
homeownership_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
houstorepair	267	0.97	9.900000e-01	0.02	5.2000e-01	0.98	0.99	1.000000e+00	1.000000e+00	
houstorepair_pctile	267	0.97	5.000000e-01	0.28	0.0000e+00	0.25	0.50	8.100000e-01	8.100000e-01	
inhighschool	267	0.97	9.800000e-01	0.07	0.0000e+00	0.99	1.00	1.000000e+00	1.000000e+00	
inhighschool_pctile	267	0.97	5.000000e-01	0.22	0.0000e+00	0.25	0.63	6.300000e-01	6.300000e-01	
inpreschool	267	0.97	5.300000e-01	0.28	0.0000e+00	0.33	0.52	7.300000e-01	1.000000e+00	
inpreschool_pctile	267	0.97	5.000000e-01	0.29	3.0000e-02	0.25	0.50	7.500000e-01	9.500000e-01	
insured	267	0.97	8.900000e-01	0.08	2.6000e-01	0.85	0.91	9.500000e-01	1.000000e+00	
insured_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
ownsevere	267	0.97	1.200000e-01	0.08	0.0000e+00	0.08	0.11	1.500000e-01	1.000000e+00	
ownsevere_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	9.900000e-01	
ozone	267	0.97	5.000000e-02	0.01	3.0000e-02	0.04	0.05	6.000000e-02	7.000000e-02	
ozone_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.400000e-01	1.000000e+00	
parkaccess	267	0.97	7.700000e-01	0.31	0.0000e+00	0.60	0.94	1.000000e+00	1.000000e+00	
parkaccess_pctile	267	0.97	5.000000e-01	0.28	2.0000e-02	0.25	0.50	8.000000e-01	8.000000e-01	
percapitaincome	267	0.97	3.806794e+02	2284.93	8.0790e+02	2092.00	32317.50	4.759150e+04	1.743880e+05	
percapitaincome_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
pm25	267	0.97	1.018000e+01	1.15	1.8800e+00	8.59	10.17	1.194000e+01	1.639000e+01	
pm25_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
rentsevere	267	0.97	2.500000e-01	0.11	0.0000e+00	0.18	0.25	3.200000e-01	7.500000e-01	
rentsevere_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	9.900000e-01	
retail	267	0.97	6.960000e+00	20.84	0.0000e+00	0.97	2.75	6.770000e+00	8.025300e+02	
retail_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
treecanopy	267	0.97	8.000000e-02	0.09	0.0000e+00	0.04	0.05	9.000000e-02	7.400000e-01	
treecanopy_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
uncrowded	267	0.97	9.100000e-01	0.09	3.3000e-01	0.87	0.94	9.800000e-01	1.000000e+00	
uncrowded_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	9.700000e-01	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
voting	267	0.97	7.700000e-01	0.10	3.1000e-01	0.71	0.79	8.500000e-01	9.600000e-01	
voting_pctile	267	0.97	5.000000e-01	0.29	0.0000e+00	0.25	0.50	7.500000e-01	1.000000e+00	
latino_pct	45	0.99	3.800000e-01	0.26	0.0000e+00	0.16	0.31	5.800000e-01	1.000000e+00	
white_pct	45	0.99	3.900000e-01	0.26	0.0000e+00	0.15	0.37	6.100000e-01	1.000000e+00	
black_pct	45	0.99	6.000000e-02	0.09	0.0000e+00	0.01	0.03	7.000000e-02	8.500000e-01	
asian_pct	45	0.99	1.400000e-01	0.16	0.0000e+00	0.03	0.08	1.800000e-01	9.400000e-01	
multiple_pct	45	0.99	3.000000e-02	0.02	0.0000e+00	0.01	0.03	4.000000e-02	1.700000e-01	
native_am_pct	45	0.99	0.000000e+00	0.02	0.0000e+00	0.00	0.00	0.000000e+00	1.000000e+00	
pacific_isl_pct	45	0.99	0.000000e+00	0.01	0.0000e+00	0.00	0.00	0.000000e+00	1.000000e+00	
other_pct	45	0.99	0.000000e+00	0.01	0.0000e+00	0.00	0.00	0.000000e+00	1.000000e+00	

Variable type: POSIXct

skim_variable	n_missing	complete_rate	min	max	median	n_unique
version	0	1	2022-02-10	2022-02-10	2022-02-10	1

Check missing counts across all columns

```
hpi_missing <- hpi %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "n_missing") %>%
  filter(n_missing > 0) %>%
  arrange(desc(n_missing))

print(hpi_missing)
```

```
## # A tibble: 79 x 2
##   variable      n_missing
##   <chr>         <int>
## 1 notes          8056
## 2 leb            267
## 3 leb_pctile     267
## 4 hpi            267
## 5 hpi_pctile     267
## 6 hpi_quartile   267
## 7 hpi_least_healthy_25pct 267
## 8 economic       267
## 9 economic_pctile 267
```

```
## 10 education
## # i 69 more rows
```

267

4. CLEAN HPI DATASET

```
hpi_clean <- hpi %>%

# --- 4a. Drop metadata / admin columns not useful for modeling ---
select(-version, -notes) %>%

# --- 4b. Ensure GEO_ID is character (11-digit FIPS, zero-padded) ---
# HPI stores CA tracts as 10-digit integers; pad to standard 11-digit FIPS
mutate(
  geo_id = str_pad(as.character(geo_id), width = 11, side = "left", pad = "0")
) %>%

# --- 4c. Fix UrbanType: "no_data" string → true NA ---
mutate(
  urban_type = na_if(urban_type, "no_data"),
  urban_type = factor(urban_type,
    levels = c("rural", "urban_cluster", "urban_area"))
) %>%

# --- 4d. Remove tracts with zero population (no meaningful health data) ---
filter(pop > 0) %>%

# --- 4e. Flag and optionally remove tracts missing LEB / HPI ---
# These 267 tracts are missing both - likely suppressed by CDC for small pop.
# We keep them in a separate object for transparency; drop from model data.
mutate(missing_leb = is.na(leb))

# Separate: tracts with and without life expectancy
hpi_model <- hpi_clean %>% filter(!missing_leb) # ~7,790 tracts for modeling
hpi_no_leb <- hpi_clean %>% filter(missing_leb) # 267 tracts - inspect later

cat("Tracts available for modeling:", nrow(hpi_model), "\n")

## Tracts available for modeling: 7790

cat("Tracts dropped (missing LEB):", nrow(hpi_no_leb), "\n")

## Tracts dropped (missing LEB): 222

# --- 4f. Sanity-check numeric ranges for key indicators ---
# Percentile columns should be [0, 1] or [0, 100] - confirm
hpi_model %>%
```



```
select(ends_with("_pctile")) %>%
  summarise(across(everything(), list(min = min, max = max), na.rm = TRUE)) %>%
  glimpse()
```

```
## Rows: 1
## Columns: 66
## $ leb_pctile_min          <dbl> 0.00012837
## $ leb_pctile_max          <dbl> 1
## $ hpi_pctile_min          <dbl> 0.00012837
## $ hpi_pctile_max          <dbl> 1
## $ economic_pctile_min     <dbl> 0.00012837
## $ economic_pctile_max     <dbl> 1
## $ education_pctile_min    <dbl> 0.00012837
## $ education_pctile_max    <dbl> 1
## $ insurance_pctile_min    <dbl> 0.00012837
## $ insurance_pctile_max    <dbl> 0.9982028
## $ clean_enviro_pctile_min <dbl> 0.00012837
## $ clean_enviro_pctile_max <dbl> 1
## $ housing_pctile_min      <dbl> 0.00012837
## $ housing_pctile_max      <dbl> 1
## $ neighborhood_pctile_min <dbl> 0.00012837
## $ neighborhood_pctile_max <dbl> 1
## $ social_pctile_min       <dbl> 0.00012837
## $ social_pctile_max       <dbl> 1
## $ transportation_pctile_min <dbl> 0.00012837
## $ transportation_pctile_max <dbl> 1
## $ abovepoverty_pctile_min <dbl> 0.00012837
## $ abovepoverty_pctile_max <dbl> 1
## $ automobile_pctile_min   <dbl> 0.00012837
## $ automobile_pctile_max   <dbl> 0.9827985
## $ bachelorsed_pctile_min  <dbl> 0.00012837
## $ bachelorsed_pctile_max  <dbl> 1
## $ censusresponse_pctile_min <dbl> 0.00012837
## $ censusresponse_pctile_max <dbl> 1
## $ commute_pctile_min      <dbl> 0.01489089
## $ commute_pctile_max      <dbl> 1
## $ dieselpm_pctile_min     <dbl> 0.00012837
## $ dieselpm_pctile_max     <dbl> 1
## $ employed_pctile_min     <dbl> 0.00012837
## $ employed_pctile_max     <dbl> 1
## $ h20contam_pctile_min    <dbl> 0.00012837
## $ h20contam_pctile_max    <dbl> 0.9995507
## $ homeownership_pctile_min <dbl> 0.0021181
## $ homeownership_pctile_max <dbl> 1
## $ houserepair_pctile_min  <dbl> 0.00012837
## $ houserepair_pctile_max  <dbl> 0.8094352
## $ inhighschool_pctile_min <dbl> 0.000320924
## $ inhighschool_pctile_max <dbl> 0.6301669
## $ inpreschool_pctile_min  <dbl> 0.02535302
## $ inpreschool_pctile_max  <dbl> 0.9508986
## $ insured_pctile_min      <dbl> 0.00012837
## $ insured_pctile_max      <dbl> 0.9982028
## $ ownsevere_pctile_min    <dbl> 0.000641849
```

```
## $ ownsevere_pctile_max      <dbl> 0.9922978
## $ ozone_pctile_min         <dbl> 0.004685494
## $ ozone_pctile_max         <dbl> 0.9991656
## $ parkaccess_pctile_min     <dbl> 0.01655969
## $ parkaccess_pctile_max     <dbl> 0.8005135
## $ percapitaincome_pctile_min <dbl> 0.00012837
## $ percapitaincome_pctile_max <dbl> 1
## $ pm25_pctile_min          <dbl> 0.00012837
## $ pm25_pctile_max          <dbl> 1
## $ rentsevere_pctile_min     <dbl> 0.00012837
## $ rentsevere_pctile_max     <dbl> 0.9915918
## $ retail_pctile_min         <dbl> 0.000320924
## $ retail_pctile_max         <dbl> 1
## $ treecanopy_pctile_min     <dbl> 0.00012837
## $ treecanopy_pctile_max     <dbl> 1
## $ uncrowded_pctile_min      <dbl> 0.00012837
## $ uncrowded_pctile_max      <dbl> 0.9743902
## $ voting_pctile_min         <dbl> 0.00012837
## $ voting_pctile_max         <dbl> 1
```

```
# Race/ethnicity pcts should sum to ~100 per tract
hpi_model <- hpi_model %>%
  mutate(
    pct_sum = latino_pct + white_pct + black_pct + asian_pct +
              multiple_pct + native_am_pct + pacific_isl_pct + other_pct
  )

# Flag tracts where race pcts don't sum to ~100 (± 2 pp rounding tolerance)
race_check <- hpi_model %>%
  filter(abs(pct_sum - 100) > 2) %>%
  select(geo_id, name, pct_sum)

cat("Tracts with race pct sum outside 98-102%:", nrow(race_check), "\n")
```

```
## Tracts with race pct sum outside 98-102%: 7790
```

```
# Drop the diagnostic column
hpi_model <- hpi_model %>% select(-pct_sum, -missing_leb)
```

5. CLEAN USALEEP DATASET

CA_A_CSV.xlsx columns after clean_names():

tract_id, state2kx, cnty2kx, tract2kx, e_0_, se_e_0_, abridged_life_table_flag

File is already California-only — no state filtering needed.

Flag values: 1 = <25 deaths (least reliable), 2 = 25–99, 3 = 100+ deaths

```

usaleep_clean <- usaleep %>%
  rename(
    geo_id   = tract_id,
    leb_cdc  = e_0,
    leb_se   = se_e_0,
    leb_flag = abridged_life_table_flag
  ) %>%
  mutate(
    geo_id = str_pad(as.character(geo_id), width = 11, side = "left", pad = "0")
  ) %>%
  # Drop flag 1 tracts (fewer than 25 deaths - estimates least reliable)
  filter(leb_flag != 1) %>%
  select(geo_id, leb_cdc, leb_se, leb_flag)

cat("CA tracts in USALEEP after filtering:", nrow(usaleep_clean), "\n")

## CA tracts in USALEEP after filtering: 7100

# Expect ~7,100 tracts after dropping flag 1 (~416 removed)

```

6. JOIN HPI + USALEEP ON CENSUS TRACT ID

Note: HPI LEB is derived from CA-specific data; USALEEP is the CDC source.

We use USALEEP as our official response variable (leb_cdc).

The HPI's own LEB (leb) is kept for comparison / model benchmarking.

```

merged <- hpi_model %>%
  left_join(usaleep_clean, by = "geo_id")

cat("Tracts after join:", nrow(merged), "\n")

## Tracts after join: 7790

cat("Tracts missing CDC LEB after join:", sum(is.na(merged$leb_cdc)), "\n")

## Tracts missing CDC LEB after join: 776

# Drop tracts that couldn't be matched or are missing CDC life expectancy
final_df <- merged %>%
  filter(!is.na(leb_cdc))

cat("Final analysis-ready tracts:", nrow(final_df), "\n")

## Final analysis-ready tracts: 7014

```

7. CREATE DERIVED / HELPER VARIABLES

```
final_df <- final_df %>%  
  mutate(  
    # Income quartile based on percapitaincome (useful for 2-way ANOVA)  
    income_quartile = ntile(percapitaincome, 4),  
    income_quartile = factor(income_quartile,  
                             labels = c("Q1_low", "Q2", "Q3", "Q4_high")),  
  
    # Majority-minority flag (non-white > 50%)  
    majority_minority = if_else(white_pct < 50, 1L, 0L),  
  
    # HPI quartile as factor (already exists; ensure it's a factor)  
    hpi_quartile = factor(hpi_quartile),  
  
    # Urban type with explicit NA as "unknown" level for modeling  
    urban_type = fct_explicit_na(urban_type, na_level = "unknown")  
  )
```

8. VARIABLE GROUPS

```
# Raw indicator variables (23 HPI indicators, no percentile versions)  
indicators_raw <- c(  
  "abovepoverty", "automobile", "bachelors", "censusresponse",  
  "commute", "dieselpm", "employed", "h2ocontam", "homeownership",  
  "housethree", "inhighschool", "inpreschool", "insured",  
  "ownsevere", "ozone", "parkaccess", "percapitaincome",  
  "pm25", "rentsevere", "retail", "treecanopy", "uncrowded", "voting"  
)  
  
# Domain composite scores  
domain_scores <- c(  
  "economic", "education", "insurance", "clean_enviro",  
  "housing", "neighborhood", "social", "transportation"  
)  
  
# Demographic variables  
demo_vars <- c(  
  "pop", "pctgqtract",  
  "latino_pct", "white_pct", "black_pct", "asian_pct",  
  "multiple_pct", "native_am_pct", "pacific_isl_pct", "other_pct"  
)  
  
# Response variable  
response_var <- "leb_cdc"
```

9. SPLIT INTO MODELING SUBSETS

```
# 80/20 train-test split (stratified by income_quartile)

split      <- initial_split(final_df, prop = 0.80, strata = income_quartile)
train_data <- training(split)
test_data  <- testing(split)

cat("Training rows:", nrow(train_data), "\n")
```

```
## Training rows: 5610
```

```
cat("Test rows:      ", nrow(test_data), "\n")
```

```
## Test rows:      1404
```

10. SAVE CLEANED DATA

```
write_csv(final_df, "hpi_clean_full.csv")
write_csv(train_data, "hpi_train.csv")
write_csv(test_data, "hpi_test.csv")

# Save variable group vectors for use in modeling scripts
saveRDS(
  list(
    indicators_raw = indicators_raw,
    domain_scores = domain_scores,
    demo_vars     = demo_vars,
    response_var  = response_var
  ),
  file = "variable_groups.rds"
)

cat("\n Cleaning complete. Files written:\n")
```

```
##
## Cleaning complete. Files written:
```

```
cat(" hpi_clean_full.csv  -", nrow(final_df), "tracts,", ncol(final_df), "cols\n")
```

```
## hpi_clean_full.csv - 7014 tracts, 87 cols
```

```
cat(" hpi_train.csv      -", nrow(train_data), "tracts\n")
```

```
## hpi_train.csv       - 5610 tracts
```

```
cat(" hpi_test.csv       -", nrow(test_data), "tracts\n")
```

```
## hpi_test.csv        - 1404 tracts
```

```
cat(" variable_groups.rds - variable name lists for modeling\n")
```

```
## variable_groups.rds - variable name lists for modeling
```